

Graph-Aware Decentralized Burst-Split Control for Dynamic Clustered FANETs

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Abstract—Flying ad hoc networks (FANETs) require communication-control policies that can react to rapid topology variation, uneven traffic load, spatially varying interference, and local coordinator failures. Existing global MAC-selection or static scheduling abstractions are often too coarse for clustered FANETs because they cannot distinguish local intra-cluster service demand from inter-cluster coordination demand. This paper formulates FANET communication control as a decentralized graph-coupled decision problem in which cluster-heads act as local agents over a time-varying cluster interference graph. Each cluster-head jointly selects its access mode and a burst-split ratio that allocates burst time between intra-cluster service and inter-cluster coordination. To solve the resulting problem, we develop MAGAT-D3QN, a graph-attention recurrent dueling multi-agent reinforcement-learning controller, and compare it with MAPPO, VDN, QMIX, IQL, fixed/heuristic burst-allocation policies under common cluster-level action semantics. Across the evaluated simulator regimes, MAGAT-D3QN achieves competitive throughput while eliminating packet drops and collisions at the nominal load, improves learned burst allocation over fixed and heuristic policies, and maintains robust behavior under topology variation, leader failures, train-test mismatch, imperfect graph information, and hardware-oriented profiling. The results support a reliability- and coordination-oriented advantage rather than universal throughput or delay dominance.

Index Terms—Flying ad hoc networks, decentralized control, dynamic clustering, graph attention networks, multi-agent reinforcement learning, burst allocation, fault tolerance.

I. INTRODUCTION

FLYING ad hoc networks (FANETs) must support sensing, relaying, and mission-critical information exchange under rapid topology variation, intermittent connectivity, spatially varying interference, and nonuniform traffic demand [1]–[5]. Unlike many terrestrial wireless networks, FANETs operate in a regime where mobility, queue evolution, interference coupling, and coordination overhead evolve on comparable time scales. Communication control in such networks therefore requires policies that are decentralized, topology-sensitive, and robust to local coordinator disruptions.

A common abstraction is to treat FANET communication control as a global MAC-selection problem, where one network-wide decision chooses a protocol mode at each

epoch [6]–[8]. This abstraction is weak in clustered FANETs. Spatially separated clusters may simultaneously experience different backlog levels, relay burden, interference exposure, and coordination demand. A single global mode suppresses this heterogeneity and, more importantly, cannot represent the tradeoff between intra-cluster service and inter-cluster coordination when both share the same limited burst budget.

This paper formulates communication control at the cluster-head level. Let the swarm of n UAVs be partitioned into $c(t)$ dynamic clusters $\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} C_k(t)$, where each cluster $C_k(t)$ is represented by a cluster-head $L_k(t)$. The active cluster-heads interact over a time-varying interference graph $G_t = (V_t, E_t)$ whose edges capture inter-cluster coupling through contention, relaying, and coordination backlog. Each cluster-head selects the local action

$$a_k(t) = (m_k(t), \rho_k(t)), \quad (1)$$

where $m_k(t)$ is the local access mode and $\rho_k(t)$ is a burst-split ratio. The burst duration t_{burst} is divided into an intra-cluster service phase $T_{1,k}(t) = \rho_k(t)t_{\text{burst}}$ and an inter-cluster coordination phase $T_{2,k}(t) = (1 - \rho_k(t))t_{\text{burst}}$. Thus, the controller must learn not only which local access behavior to use, but also how much burst time to assign to service versus coordination as cluster size, backlog, graph degree, and leader-health conditions change.

The resulting problem is a decentralized graph-coupled control process. Each cluster-head acts using local observations and neighborhood summaries, while the transition dynamics remain coupled through UAV mobility, interference, reassociation, split/merge events, and local leader handover. To address this problem, we develop a graph-aware multi-agent reinforcement-learning controller, MAGAT-D3QN, that combines graph attention, recurrent memory, and dueling action evaluation under decentralized execution. The method is compared with representative MARL baselines, including MAPPO, VDN, QMIX, and IQL, together with fixed and heuristic burst-allocation policies and structural ablations under common cluster-level decision semantics.

The main contributions are as follows:

- 1) We formulate dynamic clustered FANET communication control as a decentralized graph-coupled decision problem in which cluster-heads are the agents and the network evolves through reassociation, split/merge events, and local leader handover.
- 2) We introduce an explicit burst-split action variable $\rho_k(t)$ that allocates each cluster burst between intra-cluster ser-

vice and inter-cluster coordination, exposing a tradeoff that is not captured by global MAC selection.

- 3) We develop MAGAT-D3QN, a graph-attention recurrent dueling MARL controller for cluster-head-level execution, and compare it with MAPPO, VDN, QMIX, IQL, fixed/heuristic burst-allocation policies, and architecture ablations.
- 4) We evaluate the framework through offered-load benchmarking, burst-split analysis, matched-load topology tests, cluster-dynamics sanity checks, leader-failure recovery, train-test mismatch, imperfect graph information, structural ablations, complexity analysis, and hardware-oriented profiling.

The claims are intentionally conservative. The proposed controller is not uniformly best across all metrics: MAPPO attains slightly higher raw throughput and IQL attains lower delay in the nominal benchmark. The value of MAGAT-D3QN is instead its reliability- and coordination-oriented tradeoff: it achieves competitive throughput while eliminating packet drops and collisions at the nominal load, and its ablations show that graph representation, recurrent memory, and learned burst allocation contribute to fairness, coordination stability, and robustness. The remainder of the paper presents the related work, formulation, proposed method, experimental protocol, results, limitations, and conclusions.

II. RELATED WORK

Decentralized wireless scheduling and MAC. Classical wireless scheduling and distributed MAC mechanisms provide contention resolution and queue-aware resource allocation under partial information [6]–[10]. These methods are scalable and analytically well motivated, but their decision granularity is usually link-, node-, or network-mode based. They are not designed to represent dynamic coordination among clustered subnetworks whose local service and inter-cluster signaling demands evolve differently over time. Recent learning-based methods improve adaptability for spectrum access, power control, and wireless scheduling [11]–[14]; however, many such formulations still operate on flat node sets or global control abstractions that do not explicitly model the service-coordination split inside clustered FANET operation.

Graph learning for wireless control. Graph neural networks (GNNs) provide a natural representation for wireless systems because interference, contention, and neighborhood coupling are sparse and relational [15]–[17]. They have been used for wireless resource allocation [18], power control [19], [20], heterogeneous scheduling [21], [22], and multi-UAV D2D communication [23]. Graph attention is especially useful when neighboring entities exert unequal influence [16], [24]. Nevertheless, existing graph-learning wireless controllers usually define graph nodes as links, radios, users, or generic embeddings. In contrast, the present work defines graph nodes as dynamic cluster-heads that are also the decentralized decision makers, so the graph is not only a feature extractor but the interaction structure of the control problem itself.

MARL in UAV/FANET communication. Multi-agent reinforcement learning has been widely studied for partially

observable and nonstationary wireless systems [25]–[27]. Value-decomposition methods such as VDN [28] and QMIX [29], independent-learning methods such as IQL, and policy-gradient methods such as MAPPO [30] provide relevant baselines for cooperative multi-agent communication control. In UAV and FANET settings, learning-based approaches have addressed trajectory–resource co-design [31], cooperative multi-UAV networking [32], swarm communication [33], joint communication–computing control [34], DRL-based routing [35], and cellular-connected UAV navigation [36]. Surveys further summarize UAV communication architectures and open challenges [3]–[5]. However, most of these studies use flat agent sets, trajectory/resource actions, routing actions, or homogeneous communication abstractions, rather than cluster-head agents that jointly select access behavior and burst allocation over a time-varying inter-cluster graph.

Clustered FANET and fault-tolerant control. Clustering is a standard mechanism for improving scalability and reducing control overhead in mobile ad hoc and aerial networks [37]–[41]. Recent FANET and UAV-network studies have considered adaptive clustering, cluster-based routing, intelligent transportation clustering, fault-tolerant cluster-head election, and cooperative recovery mechanisms [42]–[46]. Fairness across distributed users and resilience in emergency UAV networks are also relevant system-level concerns [47], [48]. These works establish the importance of hierarchy and robustness, but they primarily focus on formation, election, routing, or recovery. They do not formulate burst-level communication management as a decentralized graph-coupled control problem among cluster-heads.

Gap and positioning. Existing work does not jointly provide: i) dynamic cluster-heads as the decentralized agent set; ii) a time-varying inter-cluster interference graph as the control-coupling structure; and iii) a local action that jointly selects access mode and burst split between intra-cluster service and inter-cluster coordination. This paper addresses that intersection by combining dynamic clustering, graph-aware MARL, adaptive burst allocation, local leader handover, and robustness-oriented evaluation under topology variation, train-test mismatch, imperfect graph information, and hardware-oriented profiling.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Clustered FANET Model

Consider a FANET of n UAVs partitioned at decision instant t into $c(t)$ active clusters:

$$\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} C_k(t), \quad C_k(t) \cap C_\ell(t) = \emptyset, \quad k \neq \ell. \quad (2)$$

Each cluster $C_k(t)$ has one cluster-head $L_k(t) \in C_k(t)$, which acts as the decentralized decision-making agent. Subordinate UAVs do not select communication-control actions directly; their queue, mobility, and link states are summarized in the corresponding cluster-head observation.

Cluster membership is governed by a composite score that reflects proximity, link quality, mobility compatibility, and

load balance. Reassociation from the current cluster u to a candidate cluster v is allowed only when

$$S_{i,v}(t) - S_{i,u}(t) \geq \Delta_{\text{reasoc}}, \quad (3)$$

where $\Delta_{\text{reasoc}} > 0$ suppresses oscillatory switching. The cluster set also evolves through split events when $|C_k(t)| > N_{\text{split}}^{\text{max}}$, merge events when $|C_k(t)| < N_{\text{merge}}^{\text{min}}$, and local leader handover when the current cluster-head becomes unreliable. During handover, the replacement is selected locally as

$$L_k(t^+) = \arg \max_{u \in C_k(t) \setminus \{L_k(t)\}} H_u(t), \quad (4)$$

where $H_u(t)$ combines normalized residual energy, connectivity degree, mobility stability, and queue burden. This handover rule preserves decentralized operation because no network-wide controller is required to replace a failed cluster-head.

B. Cluster Interference Graph

The interaction among active cluster-heads is represented by a time-varying cluster interference graph

$$G_t = (V_t, E_t), \quad V_t = \{1, \dots, c(t)\}, \quad (5)$$

where vertex k corresponds to cluster-head $L_k(t)$. An undirected edge $(k, \ell) \in E_t$ exists when

$$d_{k\ell}(t) \leq R_I \quad \text{or} \quad \hat{\gamma}_{k\ell}(t) \leq \tau_I, \quad (6)$$

where $d_{k\ell}(t)$ is the inter-head distance, R_I is the inter-cluster interaction radius, $\hat{\gamma}_{k\ell}(t)$ is an interference or link-quality proxy, and τ_I is the interaction threshold. The graph G_t is therefore the structural object through which interference, relaying pressure, and coordination coupling enter the decentralized control problem.

C. Local Observation and Burst-Split Action

Each decision step corresponds to one communication burst of duration t_{burst} . Cluster-head k acts from a local observation

$$\mathbf{o}_k(t) = [\phi_k^{\text{loc}}(t), \phi_k^{\text{graph}}(t), \phi_k^{\text{queue}}(t), \phi_k^{\text{health}}(t), \phi_k^{\text{hist}}(t)], \quad (7)$$

which summarizes local cluster state, graph-neighbor information, backlog and coordination demand, leader-health indicators, and recent burst-split history. The observation contains no global network state.

The local action is

$$a_k(t) = (m_k(t), \rho_k(t)), \quad (8)$$

where $m_k(t) \in \{0:\text{TDMA}, 1:\text{CSMA/CA}\}$ is the local access mode and $\rho_k(t) \in [0, 1]$ is the burst-split ratio. The burst is divided as

$$T_{1,k}(t) = \rho_k(t)t_{\text{burst}}, \quad (9)$$

$$T_{2,k}(t) = (1 - \rho_k(t))t_{\text{burst}}, \quad (10)$$

where $T_{1,k}(t)$ is used for intra-cluster head-subordinate service and $T_{2,k}(t)$ is used for inter-cluster head-head coordination. Feasible actions satisfy

$$T_{1,k}(t) \geq T_1^{\text{min}}, \quad T_{2,k}(t) \geq T_2^{\text{min}}. \quad (11)$$

The decentralized network action is $a(t) = \{a_k(t)\}_{k=1}^{c(t)}$; no single network-wide MAC action replaces these local decisions.

D. Reward and Objective

The local reward for cluster-head k at step t is

$$\begin{aligned} r_k(t) = & \alpha_1 \tilde{T}_k^{\text{intra}}(t) + \alpha_2 \tilde{T}_k^{\text{inter}}(t) - \beta_1 \tilde{D}_k^{\text{intra}}(t) - \beta_2 \tilde{D}_k^{\text{inter}}(t) \\ & - \eta_1 \tilde{C}_k(t) - \eta_2 \tilde{I}_k(t) - \eta_3 \tilde{Q}_k(t) \\ & - \eta_4 \tilde{E}_k(t) - \eta_5 \tilde{F}_k(t). \end{aligned} \quad (12)$$

where $\tilde{(\cdot)}$ denotes normalized quantities clipped to $[0, 1]$. The positive terms reward intra-cluster and inter-cluster throughput, while the penalty terms account for delay, collisions, interference, queue stress, energy cost, and failure-related disruption. The normalization constants and reward weights are specified in Section V.

For team-level monitoring and optional regularization, we define

$$R_{\text{team}}(t) = \frac{1}{c(t)} \sum_{k=1}^{c(t)} r_k(t) + \lambda_J J(t), \quad (13)$$

where $\lambda_J \geq 0$ weights Jain's fairness index

$$J(t) = \frac{\left(\sum_{k=1}^{c(t)} x_k(t)\right)^2}{c(t) \sum_{k=1}^{c(t)} x_k^2(t)}, \quad (14)$$

computed over nonnegative per-cluster throughput values $x_k(t)$.

E. Dec-POMDP Interpretation

The above formulation defines a decentralized partially observable Markov decision process (Dec-POMDP). A Markov decision process assumes that the next network state depends on the present state and selected actions, rather than on the full past trajectory. In a partially observable setting, however, an individual controller cannot directly observe the complete network state. This is the case here: cluster-head k observes only its local cluster state, neighborhood summaries, queue indicators, leader-health information, and recent burst history, not the full swarm state.

Each cluster-head therefore selects

$$\begin{aligned} a_k(t) & \sim \pi_{\theta,k}(\cdot | \tau_k(t)), \\ \tau_k(t) & = \{\mathbf{o}_k(0), \dots, \mathbf{o}_k(t)\}, \end{aligned} \quad (15)$$

where $\tau_k(t)$ is the local observation history used to compensate for partial observability. The process is decentralized because the joint action is formed by the collection of local cluster-head actions, $a(t) = \{a_k(t)\}_{k=1}^{c(t)}$, rather than by a single network-wide controller. The transition dynamics remain coupled through UAV mobility, queue evolution, interference, reassociation, split/merge events, graph evolution, and local leader handover. Thus, each cluster-head acts locally, but its action can affect neighboring clusters through contention and coordination over G_t .

IV. PROPOSED METHOD

This section presents the graph-aware decentralized controller used by each active cluster-head. The design follows the formulation in Section III: cluster-heads are the agents, the graph $G_t = (V_t, E_t)$ defines inter-cluster coupling, and the local action $a_k(t) = (m_k(t), \rho_k(t))$ jointly selects access behavior and burst allocation.

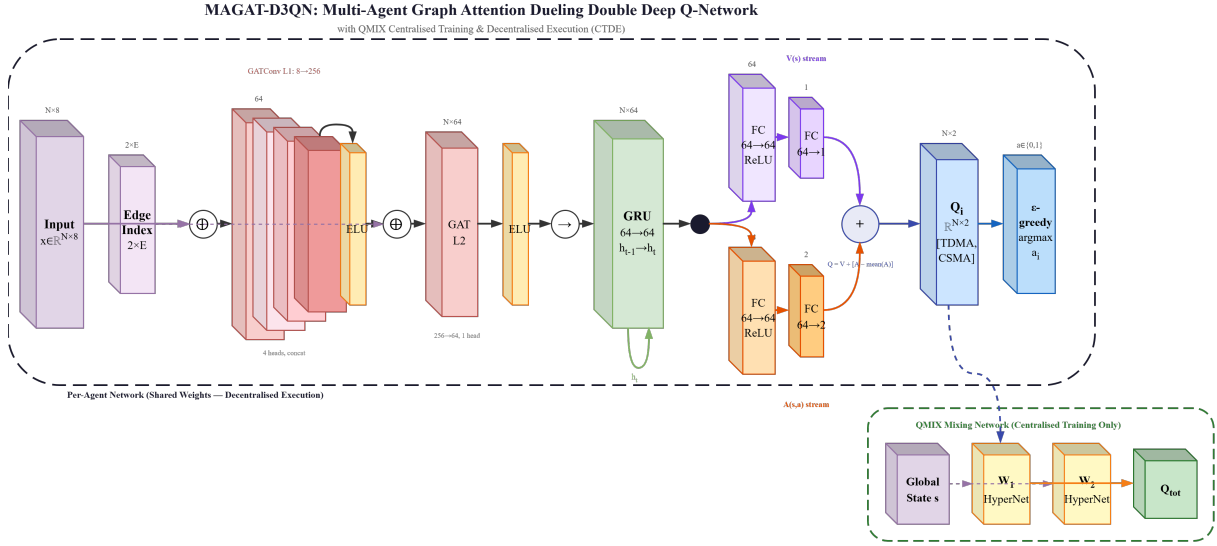


Fig. 1. Overall decentralized MAGAT-D3QN architecture. Each active cluster-head forms a local observation $o_k(t)$, exchanges graph-neighbor summaries over the cluster interference graph $G_t = (V_t, E_t)$, and uses graph attention, recurrent memory, and dueling action evaluation to select the local action $a_k(t) = (m_k(t), \rho_k(t))$. Execution is decentralized; no centralized single-agent controller or network-wide action vector is used.

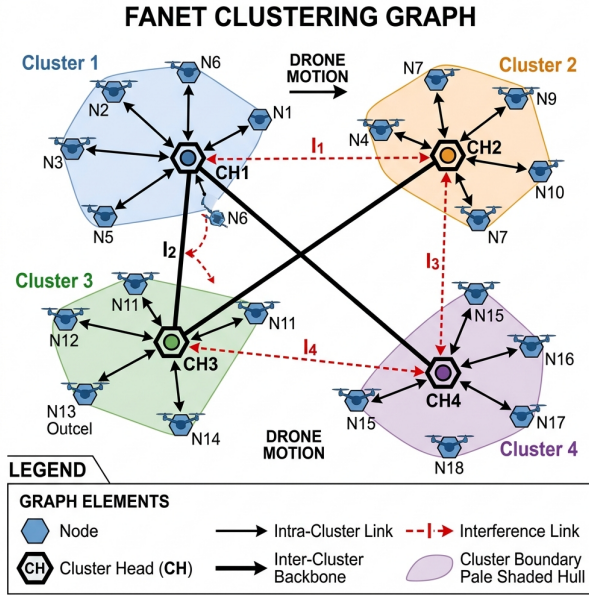


Fig. 2. Dynamic clustered FANET graph. The swarm is partitioned into clusters, each represented by a cluster-head. Active cluster-heads form the cluster interference graph $G_t = (V_t, E_t)$ used for decentralized graph-aware control.

A. Decentralized Control Pipeline

At each burst decision instant t , the environment first updates the cluster partition, local leader status, and cluster interference graph. Each active cluster-head $k \in V_t$ then forms its local observation $o_k(t)$ from cluster state, queue state, neighbor summaries, leader-health indicators, and recent burst-split history. The learned policy maps this information to a local action

$$a_k(t) = (m_k(t), \rho_k(t)), \quad k = 1, \dots, c(t), \quad (16)$$

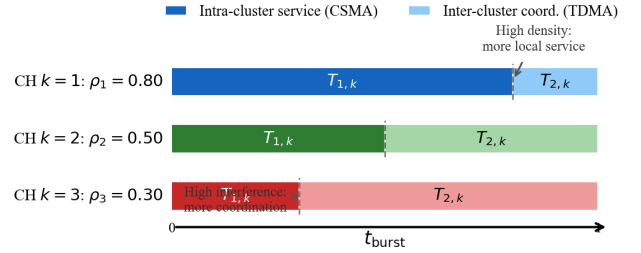


Fig. 3. Burst-split structure. Each burst of duration t_{burst} is partitioned into intra-cluster service time $T_{1,k}(t) = \rho_k(t)t_{\text{burst}}$ and inter-cluster coordination time $T_{2,k}(t) = (1 - \rho_k(t))t_{\text{burst}}$.

which is executed without any centralized action aggregator. Coupling among agents enters only through graph-neighbor information and through the inter-cluster coordination phase of the burst.

Let

$$A_{k\ell}(t) = \begin{cases} 1, & (k, \ell) \in E_t, \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

denote the cluster-head adjacency matrix, and let

$$\mathcal{N}_k(t) = \{\ell \in V_t : A_{k\ell}(t) = 1\} \quad (18)$$

be the neighbor set of cluster-head k . The graph is updated after reassociation, split, merge, or handover events so that the active node set and edge structure reflect the current clustered FANET state.

B. MAGAT-D3QN Policy

The proposed controller is a multi-agent graph-attention recurrent dueling Q-network, denoted MAGAT-D3QN. The graph-attention module encodes heterogeneous inter-cluster coupling, the recurrent block captures short-term temporal dependence under partial observability, and the dueling Q-structure improves local action ranking when several actions have similar state value.

Let $\mathbf{h}_k^{(0)}(t)$ be the initial embedding obtained from $\mathbf{o}_k(t)$. For attention layer ℓ , the unnormalized attention score from neighbor q to cluster-head k is

$$\mathbf{g}_i^{(\ell)}(t) = W^{(\ell)} \mathbf{h}_i^{(\ell)}(t), \quad (19)$$

$$e_{kq}^{(\ell)}(t) = \text{LReLU}\left(\mathbf{a}^{(\ell)\top} \left[\mathbf{g}_k^{(\ell)}(t) \parallel \mathbf{g}_q^{(\ell)}(t) \right]\right). \quad (20)$$

and the normalized attention coefficient is

$$\alpha_{kq}^{(\ell)}(t) = \frac{\exp\left(e_{kq}^{(\ell)}(t)\right)}{\sum_{j \in \mathcal{N}_k(t) \cup \{k\}} \exp\left(e_{kj}^{(\ell)}(t)\right)}. \quad (21)$$

The graph-updated representation is obtained by multi-head aggregation:

$$\mathbf{h}_k^{(\ell+1)}(t) = \sigma\left(\frac{1}{H_{\text{att}}} \sum_{p=1}^{H_{\text{att}}} \sum_{q \in \mathcal{N}_k^+(t)} \alpha_{kq}^{(\ell,p)}(t) W^{(\ell,p)} \mathbf{h}_q^{(\ell)}(t)\right), \quad (22)$$

where H_{att} is the number of attention heads, $\sigma(\cdot)$ is a non-linear activation, and $\parallel$ denotes concatenation. The attention mechanism therefore allows the policy to weight neighboring cluster-heads according to their instantaneous relevance rather than applying uniform aggregation.

The graph-updated embedding is passed through a recurrent memory block:

$$\mathbf{z}_k(t) = \text{GRU}(\mathbf{h}_k(t), \mathbf{z}_k(t-1)), \quad (23)$$

where $\mathbf{z}_k(t)$ stores short-horizon context such as queue buildup, recent burst allocation, topology changes, and hand-over transients.

The local Q-function is computed using a dueling decomposition:

$$\bar{A}_k(\mathbf{z}_k(t)) = \frac{1}{|\mathcal{A}_k|} \sum_{u' \in \mathcal{A}_k} A_k(\mathbf{z}_k(t), u'), \quad (24)$$

$$Q_k(\mathbf{o}_k(t), u) = V_k(\mathbf{z}_k(t)) + A_k(\mathbf{z}_k(t), u) - \bar{A}_k(\mathbf{z}_k(t)), \quad (25)$$

where $V_k(\cdot)$ is the value stream, $A_k(\cdot, \cdot)$ is the advantage stream, and u is a discrete local action index.

C. Burst-Allocation Action Encoding

The local action space is the Cartesian product

$$\mathcal{A}_k = \mathcal{M}_k \times \mathcal{P}_k, \quad (26)$$

where $\mathcal{M}_k$ is the MAC-mode set and $\mathcal{P}_k$ is the feasible burst-ratio set. Each pair $(m_k(t), \rho_k(t))$ is encoded as

$$u_k(t) = \text{enc}(m_k(t), \rho_k(t)), \quad (27)$$

with inverse mapping

$$(m_k(t), \rho_k(t)) = \text{dec}(u_k(t)). \quad (28)$$

This encoding allows the value-based policy to optimize MAC selection and burst allocation jointly, while the feasibility constraint in (11) prevents actions that eliminate either intra-cluster service or inter-cluster coordination.

D. Learning Objective and Decentralized Execution

For MAGAT-D3QN, the primary learning signal is the local reward $r_k(t)$ from (12). The temporal-difference loss for cluster-head k is

$$\mathcal{L}_{Q,k}(\theta) = \mathbb{E} \left[(y_k(t) - Q_k(\mathbf{o}_k(t), u_k(t); \theta))^2 \right], \quad (29)$$

with Double-DQN target

$$y_k(t) = r_k(t) + \gamma Q_k^-(\mathbf{o}_k(t+1), u_k^*(t+1); \bar{\theta}), \quad (30)$$

where

$$u_k^*(t+1) = \arg \max_{u' \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t+1), u'; \theta), \quad (31)$$

and $\bar{\theta}$ denotes the target-network parameters.

When fairness-oriented team regularization is enabled, the training loss becomes

$$\mathcal{L}_{\text{tot}}(\theta) = \sum_{k=1}^{c(t)} \mathcal{L}_{Q,k}(\theta) + \lambda_{\text{reg}} \mathcal{L}_{\text{team}}(\theta), \quad (32)$$

where $\mathcal{L}_{\text{team}}(\theta)$ is derived from the team objective in (13). This term is used only as a regularizer and does not replace the local reward.

At execution time, each cluster-head independently selects

$$\hat{u}_k(t) = \arg \max_{u \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t), u; \theta), \quad (33)$$

and decodes the physical action as

$$\hat{a}_k(t) = \text{dec}(\hat{u}_k(t)). \quad (34)$$

Thus, execution remains cluster-head-local; no centralized single-agent controller, SARL comparator, or network-wide action vector is used anywhere in the proposed method.

E. Compared Baselines and Ablation Scope

The empirical comparison uses decentralized or multi-agent baselines under the same cluster-level action semantics. The baseline set consists of IQL, VDN, QMIX, MAPPO, fixed- ρ burst allocation, and heuristic- ρ burst allocation. Structural ablations remove individual components of MAGAT-D3QN, including graph message passing, attention, recurrent memory, and burst-history features. This comparison isolates whether the observed behavior is due to graph-aware representation, temporal memory, learned burst allocation, or the underlying cluster-head formulation.

F. Complexity and Communication Overhead

Let d_h denote the latent width and H_{att} the number of graph-attention heads. A first-order graph-message-passing cost per decision step is

$$\mathcal{C}_{\text{GAT}}(t) = \mathcal{O}(H_{\text{att}} |E_t| d_h + c(t) d_h^2), \quad (35)$$

while the recurrent and action-head costs are

$$\mathcal{C}_{\text{GRU}}(t) = \mathcal{O}(c(t) d_h^2), \quad (36)$$

$$\mathcal{C}_{\text{head}}(t) = \mathcal{O}(c(t) d_h |\mathcal{A}_k|). \quad (37)$$

Communication overhead comes from neighbor-summary exchange during the coordination phase. If b_{sum} is the directed

Algorithm 1 Decentralized Cluster Update and Burst-Control Execution

Require: UAV set $\mathcal{U}(t)$, previous clusters $\{C_k(t-1)\}$, previous leaders $\{L_k(t-1)\}$, previous graph G_{t-1} , learned Q-functions $\{Q_k\}$

Ensure: Updated clusters $\{C_k(t)\}$, leaders $\{L_k(t)\}$, graph G_t , actions $\{\hat{a}_k(t)\}$, rewards $\{r_k(t)\}$

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1: Cluster update.
2: for each UAV  $i \in \mathcal{U}(t)$  do
3:   Update assignment  $\chi_i(t)$  using (3).
4: end for
5: Form clusters  $\{C_k(t)\}$  from assignments  $\{\chi_i(t)\}$ .
6: for each active cluster  $C_k(t)$  do
7:   if  $|C_k(t)| > N_{\text{split}}^{\text{max}}$  then
8:     Split  $C_k(t)$ .
9:   end if
10:  if  $|C_k(t)| < N_{\text{merge}}^{\text{min}}$  then
11:    Merge  $C_k(t)$  with a compatible cluster.
12:  end if
13:  if  $\Omega_k^{\text{fail}}(t) = 1$  then
14:    Update  $L_k(t)$  using (4).
15:  end if
16: end for
17: Construct  $G_t = (V_t, E_t)$  using (6).
18: Decentralized burst-control execution.
19: for each cluster-head  $k \in V_t$  do
20:   Form local observation  $\mathbf{o}_k(t)$ .
21:    $\hat{u}_k(t) \leftarrow \arg \max_{u \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t), u)$ .
22:    $\hat{a}_k(t) = (m_k(t), \rho_k(t)) \leftarrow \text{dec}(\hat{u}_k(t))$ .
23:    $T_{1,k} \leftarrow \rho_k(t)t_{\text{burst}}$ ,  $T_{2,k} \leftarrow (1 - \rho_k(t))t_{\text{burst}}$ .
24: end for
25: Execute intra-cluster service over  $\{T_{1,k}\}_{k \in V_t}$ .
26: Execute inter-cluster coordination over  $\{T_{2,k}\}_{k \in V_t}$  on  $G_t$ .

27: Update queues, coordination backlogs, and rewards  $\{r_k(t)\}_{k \in V_t}$ .
28: return  $\{C_k(t)\}$ ,  $\{L_k(t)\}$ ,  $G_t$ ,  $\{\hat{a}_k(t)\}$ ,  $\{r_k(t)\}$ .

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summary payload from one cluster-head to one neighbor per burst, then

$$B_{\text{comm}}(t) = b_{\text{sum}} \sum_{k=1}^{c(t)} |\mathcal{N}_k(t)| = b_{\text{sum}} c(t) \bar{d}(t), \quad (38)$$

where $\bar{d}(t)$ is the average cluster-graph degree. These expressions are complemented in Section VI by measured latency, parameter count, memory footprint, communication overhead, and hardware profiling.

““latex ““

V. EXPERIMENTAL SETUP

This section defines the simulator, training protocol, evaluation metrics, stress tests, and profiling procedure used in the empirical study. The nominal configuration, Optuna-selected hyperparameters, training protocol, and compared methods are summarized in Tables I–II. All reported performance values

are computed over five independent evaluation seeds unless stated otherwise.

A. Simulation Environment

The simulator models a three-dimensional FANET operating over a $200 \times 200 \times 50$ m region with $n = 50$ UAVs. Each episode contains $T_{\text{ep}} = 100$ decision steps, where each step corresponds to one communication burst of duration $t_{\text{burst}} = 0.5$ s. The slot-level simulation resolution is $\Delta t_{\text{slot}} = 9 \mu\text{s}$.

PHY, channel, and traffic model: The PHY layer uses a nominal data rate of $R = 5$ Mbps with QPSK modulation, transmit power $P_{\text{tx}} = 20$ dBm, noise floor $N_0 = -80$ dBm, maximum acceptable bit-error rate $\text{BER}_{\text{max}} = 10^{-3}$, and communication range $R_{\text{comm}} = 200$ m. Path loss follows

$$\text{PL}(d) = K_{\text{PL}} d^\eta, \quad K_{\text{PL}} = 10^{-5}, \eta = 2.0. \quad (39)$$

The nominal channel regime is Rician fading, consistent with Table I; AWGN, Nakagami- m , and Rayleigh channels are evaluated separately in the environment-sensitivity study. Propagation delay is computed as $\tau_{\text{prop}} = d/c$.

Each packet has payload $L_{\text{pkt}} = 1000$ bytes. Under the nominal smooth traffic profile, packet arrivals follow a Poisson process with rate λ packets/s/UAV. The offered-load sweep uses 20 equally spaced values from 50 to 1000 pps, with nominal evaluation at $\lambda = 525$ pps. Additional traffic stress tests include bursty on–off loading and heavy-tail burst multipliers.

Queue, MAC, and drop model: Each UAV maintains a FIFO queue of capacity $Q_{\text{max}} = 100$ packets. The queue of member UAV i evolves as

$$q_i(t+1) = \text{clip}(q_i(t) + a_i(t) - s_i(t) + \delta_i^{\text{ctrl}}(t), 0, Q_{\text{max}}), \quad (40)$$

where $a_i(t)$ is the packet-arrival count, $s_i(t)$ is the number of served packets, and $\delta_i^{\text{ctrl}}(t)$ captures unserved control backpressure. The reported drop count $\bar{\Pi}_{\text{drop}}$ includes both queue-overflow drops and MAC-layer drops after retry exhaustion.

For $m_k(t) = 1$, the intra-cluster sub-phase uses a CSMA/CA sub-simulator with 802.11a-style backoff timing, RTS/CTS enabled, and one maximum retry. For $m_k(t) = 0$, TDMA assigns equal-length member slots with a $1 \mu\text{s}$ guard interval. Collisions are counted as simultaneous-transmission events detected within the CSMA/CA sub-phase.

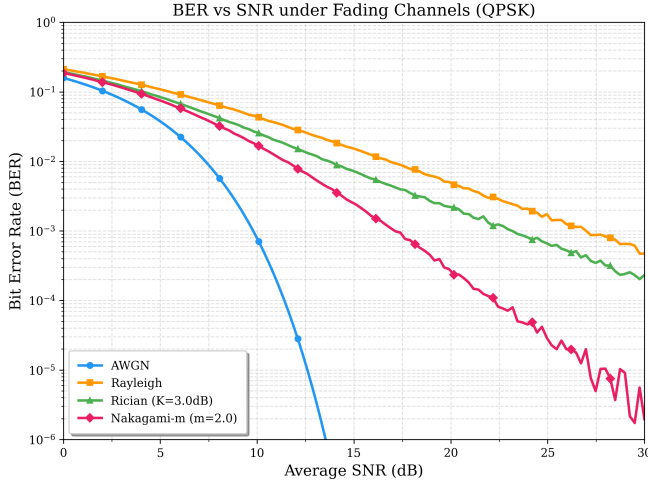
Action space and burst allocation: The local action $a_k(t) = (m_k(t), \rho_k(t))$ combines the access mode $m_k(t) \in \{0: \text{TDMA}, 1: \text{CSMA/CA}\}$ with a discretized burst-split ratio. The candidate ratio set is

$$\mathcal{P}_{\text{cand}} = \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}. \quad (41)$$

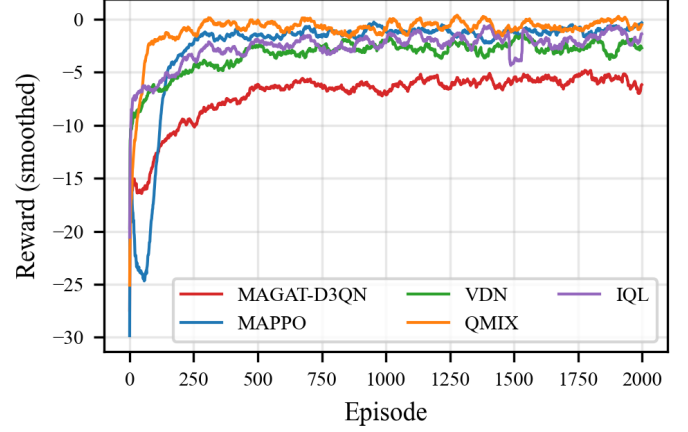
A ratio is valid when $T_{1,k} = \rho t_{\text{burst}} \geq T_1^{\text{min}}$ and $T_{2,k} = (1 - \rho)t_{\text{burst}} \geq T_2^{\text{min}}$, with $T_1^{\text{min}} = T_2^{\text{min}} = 0.05$ s. Since $t_{\text{burst}} = 0.5$ s, all nine candidates are feasible and

$$|\mathcal{A}_k| = |\mathcal{M}_k| |\mathcal{P}_k| = 2 \times 9 = 18. \quad (42)$$

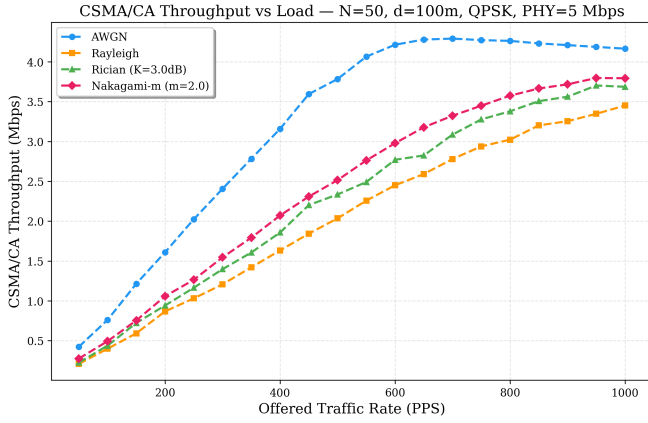
The action index is encoded as $a_{\text{id}} = m_k |\mathcal{P}_k| + \rho_{\text{id}}$ and decoded by $\rho_{\text{id}} = a_{\text{id}} \bmod |\mathcal{P}_k|$, $m_k = \lfloor a_{\text{id}} / |\mathcal{P}_k| \rfloor \bmod |\mathcal{M}_k|$.



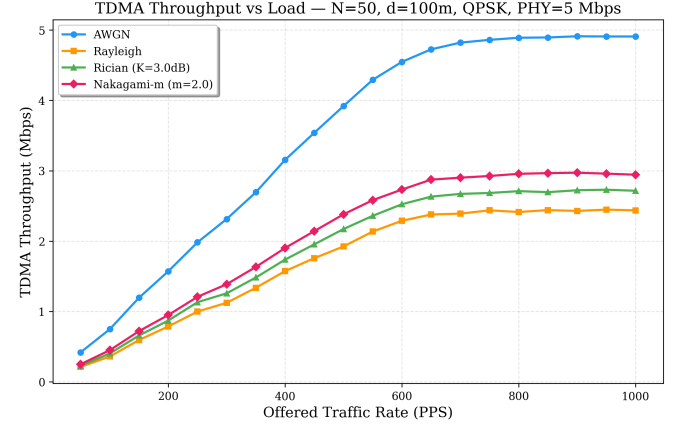
(a) BER-SNR behavior under QPSK fading channels.



(b) Training reward curves over 2000 episodes.



(c) CSMA/CA throughput versus offered load.



(d) TDMA throughput versus offered load.

Fig. 4. PHY/MAC characterization and learning convergence. (a) BER decreases with SNR under QPSK, with AWGN serving as the reference channel and Rayleigh fading showing the largest degradation. (b) Training curves verify convergence behavior over 2000 episodes; final performance claims use disjoint evaluation seeds and numerical result tables. (c) CSMA/CA throughput increases with offered load before saturation, with fading severity reducing achievable throughput. (d) TDMA shows channel-dependent throughput degradation with scheduled-access saturation behavior.

Inter-cluster coordination and interference: Neighboring cluster-heads coordinate through a tail-aligned synchronization model with guard time $\Delta t_{\text{coord}} = 0.05$ s. The temporal overlap between neighboring clusters k and j is

$$\text{overlap}(k, j) = \max(0, t_{\text{burst}} - \max(T_{1,k}, T_{1,j}) - \Delta t_{\text{coord}}), \quad (43)$$

and the coordination success ratio is

$$\text{CSR}(k, j) = \text{clip}\left(\frac{\text{overlap}(k, j)}{\zeta_{kj}}, 0, 1\right), \quad (44)$$

$$\zeta_{kj} = \max(\min(T_{2,k}, T_{2,j}) - \Delta t_{\text{coord}}, 10^{-9}).$$

The coordination, relay, and local-control capacities are $C_{\text{coord}} = 4.0$, $C_{\text{relay}} = 3.0$, and $C_{\text{local}} = 6.0$ units/s, respectively. The interference penalty for neighboring cluster-heads is $0.15(1.25 - \text{CSR})$ when both select CSMA/CA, $0.06(1.25 - \text{CSR})$ when both select TDMA, and $0.02(1.25 - \text{CSR})$ for mixed MAC modes.

Reward normalization: Reward components are clipped to $[0, 1]$ using fixed ceilings $T_{\text{max}} = 5.0$ Mbps, $D_{\text{max}} = 200$ ms,

$C_{\text{max}} = 500$ collisions, and $E_{\text{max}} = 1.0$ energy units. The implemented local reward is

$$\begin{aligned} r_k(t) = & \alpha_1 \hat{T}_k^{\text{loc}} + \alpha_2 \hat{T}_k^{\text{int}} - \beta_1 \hat{D}_k^{\text{loc}} - \beta_2 \hat{D}_k^{\text{int}} \\ & - \eta_1 \hat{C}_k - \eta_2 \hat{I}_k - \eta_3 \hat{Q}_k \\ & - \eta_4 \hat{E}_k - \eta_5^{(c)} \hat{F}_k - \eta_5^{(h)} H_k. \end{aligned} \quad (45)$$

where $H_k \in \{0, 1\}$ is the handover indicator. The default coefficients are $\alpha_1 = 0.22$, $\alpha_2 = 0.13$, $\beta_1 = 0.10$, $\beta_2 = 0.05$, $\eta_1 = 0.20$, $\eta_4 = 0.05$, $\eta_2 = 0.12$, $\eta_5^{(c)} = 0.08$, $\eta_3 = 0.10$, and $\eta_5^{(h)} = 0.05$. A Jain-fairness term with weight $\lambda_J = 0.10$ is used for team-level monitoring or regularization where applicable; it does not imply centralized execution.

Fixed and heuristic burst-allocation baselines: The heuristic burst-allocation baseline selects

$$\begin{aligned} \tilde{\rho}_k(t) &= 0.5 + 0.1 \frac{q_k(t)}{Q_{\text{max}}} - 0.05 |\mathcal{N}_k(t)|, \\ \rho_k^{\text{heur}}(t) &= \text{clip}(\tilde{\rho}_k(t), 0.10, 0.70). \end{aligned} \quad (46)$$

and discretizes the result to the nearest valid level in $\mathcal{P}_{\text{cand}}$. This rule increases intra-cluster service time under local queue pressure and increases coordination time as graph degree grows. Fixed- ρ baselines use $\rho_k(t) = 0.5$ or $\rho_k(t) = 0.8$ for all clusters and steps.

B. Failure, Mismatch, and Imperfect-Information Protocols

Leader-failure events are injected as tuples (step, target), where the target is selected from {random, max_backlog, max_degree}. Under random, the affected cluster-head is sampled uniformly from the active set; under max_backlog, the affected cluster has the largest aggregate member queue; and under max_degree, it has the largest graph degree. The event triggers local leader re-election using (4). The evaluated failure cases are single random failure, single high-backlog failure, and repeated random failures at five uniformly spaced steps.

Train-test mismatch is evaluated across node-count, mobility, interference, traffic, and topology regimes. Imperfect-information tests corrupt the observed graph $\tilde{G}_t$ through stale observations, noisy summaries, missing edges, false edges, delayed handover information, static graph, shuffled graph, and no-graph variants. Missing-edge and false-edge perturbations are applied independently per edge, while observation staleness serves $\mathbf{o}_k(t - \Delta_{\text{obs}})$ instead of $\mathbf{o}_k(t)$ and observation noise adds independent Gaussian perturbations to each feature.

C. Evaluation Metrics

For each active cluster k , throughput is $T_k(t) = B_k^{\text{local}}(t)/t_{\text{burst}}$, where $B_k^{\text{local}}(t)$ is the successfully transmitted payload in the intra-cluster sub-phase. Global throughput is

$$\bar{T} = \frac{1}{10^6} \sum_k T_k \quad (47)$$

in Mbps. Delay $\bar{D}$ is the mean end-to-end delay of successfully delivered packets, $\bar{\Pi}_{\text{drop}}$ is the total queue-overflow plus MAC-layer drop count, $\bar{C}$ is the collision count, and Jain fairness is

$$\bar{J} = \frac{(\sum_k T_k)^2}{c(t) \sum_k T_k^2}. \quad (48)$$

Episode summaries are averaged over $T_{\text{ep}} = 100$ steps and reported as mean $\pm$ standard deviation over evaluation seeds $\mathcal{S}_{\text{ev}} = \{2048, 3072, 4096, 5120, 6144\}$.

For failure recovery, the principal metrics are throughput retention, maximum delay spike, and recovery time:

$$\Gamma_{\text{ret}} = \frac{\bar{T}_{\text{failure}}}{\bar{T}_{\text{nominal}}}, \quad (49)$$

$$\Delta D_{\text{max}} = \max_{t \in [t_f, t_f + W]} D(t) - \bar{D}_{\text{nominal}}, \quad (50)$$

$$T_{\text{thr}} = (1 - \epsilon) \bar{T}_{\text{nominal}}, \quad (51)$$

$$T_{\text{rec}} = \min\{\tau > 0 : T(t_f + \tau) \geq T_{\text{thr}}\}. \quad (52)$$

For mismatch studies, relative degradation is

$$\Delta_{\text{rel}} = \frac{y_{\text{test}} - y_{\text{nominal}}}{|y_{\text{nominal}}|}, \quad (53)$$

where y is any scalar performance metric.

D. Training, Tuning, and Hardware Profiling

All learning-based controllers are trained for 2000 episodes with 100 steps per episode. Training uses seeds $\mathcal{S}_{\text{tr}} = \{42, 123, 456, 789, 1024\}$ and evaluation uses the disjoint seed set $\mathcal{S}_{\text{ev}}$. The optimizer is Adam with $(\beta_1, \beta_2) = (0.9, 0.999)$, replay size $|B| = 50,000$, hard target-network updates every 1000 steps, and an ϵ -greedy exploration schedule from 1.0 to 0.05 over 1500 episodes. Checkpoints are selected using the best mean reward over the last 100 episodes, and final evaluation uses greedy execution.

Hyperparameter selection uses Optuna with an NSGA-II sampler on the training split only. The search monitors throughput, delay, drop-rate, and collision objectives, produces a Pareto set of nondominated configurations, and selects the final reported configuration using the balanced L_2 -to-ideal rule. This prevents evaluation-seed contamination.

Hardware-oriented profiling is performed separately from simulator-side inference timing. Exported inference graphs are compiled and profiled through Qualcomm AI Hub on the Samsung Galaxy S24 Family and Dragonwing RB3 Gen 2 Vision Kit. The profiling table reports exported-graph parameter count, checkpoint size, input shape, CPU inference time, AIHub inference time, and compile/profile status. These profiling values are device-specific and are not treated as embedded-flight readiness guarantees.

VI. RESULTS AND DISCUSSION

The empirical study is organized around the reviewer questions induced by the formulation. First, we verify training and tuning behavior. Second, we evaluate nominal offered-load performance. Third, we test whether the burst-split variable is meaningful and whether topology changes induce different local decisions at matched load. Fourth, we examine cluster-dynamics sanity and leader-failure recovery. Fifth, we evaluate train-test mismatch and imperfect graph information. Finally, we isolate MAGAT-D3QN components and report software-side and hardware-oriented profiling. Unless otherwise stated, all numerical results are reported as mean $\pm$ standard deviation over five independent evaluation seeds.

A. Training and Tuning Diagnostics

Figure 4b reports the training reward traces over 2000 episodes. These curves are used only as convergence diagnostics; all performance claims are based on greedy evaluation on the disjoint evaluation seed set. Figure ?? summarizes the effect of learning rate, batch size, hidden dimension, and exploration-decay settings, while Fig. 5 shows the NSGA-II Pareto search used to select the final training configurations. The selected hyperparameters and reward weights are listed in Table II. This separation ensures that hyperparameter selection is performed on the training split and does not contaminate the final evaluation results.

B. Main Offered-Load Benchmark

The main benchmark compares MAGAT-D3QN with MAPPO, VDN, QMIX, and IQL under the same cluster-head-

TABLE I
SIMULATION, TRAINING, AND EVALUATION SETUP

Category	Configuration
Environment	200 × 200 × 50 m 3D FANET region, $n = 50$ UAVs, Random Waypoint mobility
Episode timing	$T_{\text{ep}} = 100$ decision steps, burst duration $t_{\text{burst}} = 0.5$ s, slot resolution $\Delta t_{\text{slot}} = 9 \mu\text{s}$
PHY and queue	802.11a-style PHY, 5 Mbps nominal rate, QPSK, $P_{\text{tx}} = 20$ dBm, $N_0 = -80$ dBm, $L_{\text{pkt}} = 1000$ bytes, FIFO queue capacity $Q_{\text{max}} = 100$ packets
Channel and path loss	Nominal channel: Rician fading; environment-sensitivity study includes AWGN, Rician, Nakagami- m , and Rayleigh channels; path loss $\text{PL}(d) = K_{\text{PL}} d^\eta$, $K_{\text{PL}} = 10^{-5}$, $\eta = 2.0$
Traffic	Poisson arrivals with offered-load sweep $\lambda \in [50, 1000]$ pps; nominal evaluation at $\lambda = 525$ pps; stress profiles include bursty on-off and heavy-tail traffic
Clustering and graph	Active cluster-count range 3–10; cluster update period $T_{\text{clust}} = 5$ steps; split/merge thresholds $\theta_{\text{join}} = 0.65$, $\theta_{\text{leave}} = 0.35$; graph thresholds $(R_c, R_I, \tau_I) = (100, 250, 10)$ m; handover uses health-score leader election
Action space	$m_k(t) \in \{0:\text{TDMA}, 1:\text{CSMA/CA}\}$ and $\rho_k(t) \in \{0.10, 0.20, \dots, 0.90\}$; all nine ρ levels are feasible for $t_{\text{burst}} = 0.5$ s with $T_1^{\min} = T_2^{\min} = 0.05$ s; $ \mathcal{A}_k = 2 \times 9 = 18$
Training protocol	Training seeds $\mathcal{S}_{\text{tr}} = \{42, 123, 456, 789, 1024\}$; evaluation seeds $\mathcal{S}_{\text{ev}} = \{2048, 3072, 4096, 5120, 6144\}$; Adam optimizer $(\beta_1, \beta_2) = (0.9, 0.999)$; replay buffer $ \mathcal{B} = 50,000$; hard target update every 1000 steps
Learning schedule	2000 episodes × 100 steps; ϵ -greedy exploration from 1.0 to 0.05 over 1500 episodes; checkpoint selected by best mean reward over last 100 episodes; greedy evaluation with $\epsilon = 0$
Tuning and aggregation	Optuna NSGA-II with four objectives: throughput, delay, drop rate, and collision count; final configuration selected using balanced L_2 -to-ideal rule; all reported table values are mean $\pm$ std over five evaluation seeds

TABLE II
COMPARED METHODS AND OPTUNA-SELECTED HYPERPARAMETERS

Method	Scope	Training type	Optuna-selected reward weights / rule	Architecture hyperparameters
MAGAT-D3QN	Local CH + graph	MARL	$(w_T, w_D, w_{\text{drop}}, w_{\text{col}})$ (0.15, 0.35, 0.27, 0.23)	$= h = 64, \text{lr} = 2.05 \times 10^{-4}, B = 128, \text{heads} = 2$
MAPPO	Local CH	MARL	(0.15, 0.35, 0.27, 0.23)	$h = 64, \text{lr} = 2.05 \times 10^{-4}, B = 128$
VDN	Local CH	Value decomposition	(0.16, 0.36, 0.19, 0.29)	$h = 32, \text{lr} = 1.24 \times 10^{-3}, B = 64$
QMIX	Local CH	CTDE value decomposition	(0.28, 0.03, 0.37, 0.32)	$h = 32, \text{lr} = 2.66 \times 10^{-4}, B = 128$
IQL	Local CH	Independent learning	(0.17, 0.45, 0.21, 0.17)	$h = 32, \text{lr} = 1.22 \times 10^{-3}, B = 64$

level action semantics. Figure ?? visualizes the nominal benchmark behavior, and Table III reports the aggregate statistics at $\lambda = 525$ pps.

The results show that no method dominates all metrics. MAPPO attains the highest raw throughput, 3.33 ± 0.01 Mbps, while MAGAT-D3QN achieves 3.31 ± 0.02 Mbps, which is within approximately 1.5% of the best value. IQL gives the lowest delay, 5.02 ± 0.12 ms, but has the lowest fairness, 0.822 ± 0.014 , and the highest collision count among the listed methods. MAGAT-D3QN is the only method with both zero packet drops and zero collisions while maintaining competitive throughput and fairness. Its mean delay, 9.29 ± 0.18 ms, is the highest in the table. Therefore, the correct interpretation is not uniform superiority; MAGAT-D3QN provides a reliability- and coordination-oriented tradeoff.

C. Burst-Split Behavior and Learned Allocation

The burst-split variable $\rho_k(t)$ is central to the formulation because it controls the allocation between intra-cluster service and inter-cluster coordination. Figure 6 shows the learned $\rho_k(t)$ behavior against cluster size, local backlog, graph degree, and leader-health indicators. The policy concentrates near $\rho_k(t) \approx 0.8$ during stable conditions, allocating most burst time to intra-cluster service. However, it occasionally shifts toward lower ρ values under coordination-sensitive conditions, such as congestion, handover transients, and high graph-degree interaction.

Table IV tests whether learning this variable is useful. Fixed- $\rho = 0.5$ achieves 2.07 Mbps, fixed- $\rho = 0.8$ achieves

2.89 Mbps, and heuristic- ρ achieves 2.72 Mbps. Learned- ρ achieves 3.31 Mbps and full coordination success. Thus, learned burst allocation improves throughput by approximately 60% over fixed- $\rho = 0.5$ and 14.5% over fixed- $\rho = 0.8$. The delay increases to 9.29 ms, so the improvement should be interpreted as a throughput-coordination tradeoff rather than a delay improvement.

D. Matched-Load Topology Sensitivity

To separate topology sensitivity from load sensitivity, the offered load is kept fixed while the spatial layout is varied. The three matched-load regimes are asymmetric hotspot, compact dense, and sparse separated. Figure ?? shows topology-conditioned $\rho_k(t)$ distributions, and Table V reports the corresponding throughput, delay, and fairness values.

The topology study shows that MAGAT-D3QN changes its burst-allocation behavior with graph structure. In the asymmetric-hotspot case, MAGAT-D3QN reaches 3.24 ± 0.02 Mbps and fairness 0.939 ± 0.009 . In the compact-dense case, it reaches 3.29 ± 0.02 Mbps with fairness 0.887 ± 0.010 . In the sparse-separated case, it reaches 3.17 ± 0.03 Mbps with fairness 0.848 ± 0.012 . MAPPO remains strong in raw throughput across all topologies, while QMIX shows larger fairness degradation, especially in the compact-dense and sparse-separated regimes. These results support the graph-conditioned decision claim, while also showing that topology robustness is not exclusive to MAGAT-D3QN.

E. Cluster Dynamics and Leader-Failure Recovery

Because dynamic clustering is part of the control substrate, it must remain interpretable rather than becoming a hidden source of instability. Figure 7 and Table VI report reassociation, split, merge, handover, churn, and graph-degree statistics. Under nominal load, the reassociation count is 58.8 over the 100-step episode. Under heavy-tail traffic, it increases to 2005.0, corresponding to approximately 20 individual UAV reassociation events per step for the 50-UAV swarm. This higher activity is expected because heavy-tail traffic creates rapid queue fluctuations. Importantly, the average cluster size remains bounded at 7.33 and the average graph degree remains 5.85, indicating that the clustering substrate remains structurally controlled. The compact-dense case shows similar behavior, with 1886.5 reassociations and graph degree 5.96.

Leader-failure recovery is reported in Fig. 8 and Table VII. Under single random failure, MAGAT-D3QN achieves 3.29 ± 0.02 Mbps with fairness 0.899 ± 0.009 , while MAPPO achieves 3.33 ± 0.01 Mbps with fairness 0.897 ± 0.008 . Under repeated random failures, MAGAT-D3QN maintains 3.29 ± 0.03 Mbps, but fairness decreases to 0.867 ± 0.014 . QMIX shows the largest fairness degradation in this setting, reaching 0.764 ± 0.022 . These results support the local handover mechanism under the tested failure patterns, but the claim remains conditioned on the simulated failure schedules and load regimes.

F. Generalization and Imperfect-Information Robustness

Figure 9 and Table ?? evaluate train-test mismatch across node-count, mobility, interference, and traffic regimes. For node-count, mobility, and interference mismatch, all methods retain approximately 97–101% throughput. Traffic mismatch produces the largest effect: throughput retention increases to 116–121% because the instantaneous heavy-tail load is higher, but fairness retention drops to 76–79%. This pattern is method-wide, so it should be interpreted as a regime effect rather than a MAGAT-specific weakness.

Figure 10 and Table VIII evaluate stale observations, noisy summaries, missing edges, false edges, delayed handover information, and other graph perturbations. MAGAT-D3QN retains 99% throughput and 98% fairness across the listed perturbation categories. MAPPO is also highly robust, with 100% throughput retention in the listed cases. QMIX is the most sensitive to missing-edge corruption, where its throughput retention drops to 89%. Thus, the correct claim is graceful degradation under the tested perturbations, not invariance to arbitrary graph corruption.

G. MAGAT-Isolation Ablation and Sensitivity

Table IX isolates the structural components of MAGAT-D3QN. Removing graph message passing reduces fairness from 0.912 ± 0.008 to 0.851 ± 0.018 and introduces 8 ± 3 collisions, indicating that topology-aware representation contributes primarily to fairness and coordination quality. Replacing attention with mean pooling gives fairness 0.882 ± 0.013 , suggesting that heterogeneous neighbor weighting is useful but not the only source of gain. Removing the GRU increases

collisions to 5 ± 3 and reduces fairness to 0.869 ± 0.015 , showing that temporal memory helps stabilize decisions under partial observability. Removing burst history gives a smaller degradation, with fairness 0.895 ± 0.011 . The fixed- ρ ablation reaches only 2.89 ± 0.04 Mbps, confirming that learned burst allocation contributes to throughput beyond a strong static setting.

Reward-weight sensitivity is summarized in Fig. ?? and Table ?. The throughput-heavy configuration gives the best composite score, 0.348, while delay-heavy weighting gives the lowest score, 0.047. This does not mean that throughput-heavy weighting is universally optimal; it indicates that, under the selected composite objective and simulator regime, throughput receives the strongest contribution to the aggregate score. Environment sensitivity is reported in Table X. Mean throughput retention across algorithms is 100% for AWGN, 88.3% for Rician, 90.6% for Nakagami- m , and 82.7% for Rayleigh, confirming that Rayleigh fading is the most adverse of the tested channel regimes.

““latex

H. Complexity and Hardware-Oriented Profiling

Figure ?? and Table ?? report the simulator-side inference cost. MAGAT-D3QN requires a mean inference time of 2.71 ms, with 3.15 ms and 3.57 ms at the 95th and 99th percentiles. This is higher than simpler baselines, whose mean latency ranges from about 0.35 to 0.57 ms. The increase is expected because MAGAT-D3QN performs graph-attention message passing. In terms of size, MAGAT-D3QN has 46,355 parameters and a memory footprint of 0.18 MB. This remains moderate in absolute terms, although it is larger than IQL, which has 2,450 parameters and 0.01 MB memory.

Table ?? reports Qualcomm AI Hub profiling on two device classes. On the Samsung Galaxy S24 family, all models compile and profile successfully. MAGAT-D3QN requires 56 ms inference time, which is close to IQL, MAPPO, QMIX, and VDN (49–54 ms). On the Dragonwing RB3 Gen 2 Vision Kit, MAGAT-D3QN requires 876 ms, compared with 334 ms for IQL, 252 ms for MAPPO, 270 ms for VDN, and 1663 ms for QMIX.

These results show that the proposed model can be compiled and profiled on practical hardware, but the deployment claim should be interpreted carefully. On the RB3 platform, the MAGAT-D3QN latency is higher than the default 0.5 s burst duration, so real-time use on that device would require optimization, quantization, batching changes, model simplification, or a longer control interval. On the Galaxy S24 family, the measured latency remains well below the default burst interval.

Overall, the complexity results support the main claim that graph-aware decentralized burst-split control improves coordination stability and robustness under the evaluated clustered FANET settings, while introducing a measurable but manageable computational overhead. ““

VII. CONCLUSION

This paper studied decentralized communication control for dynamic clustered FANETs. The proposed formulation

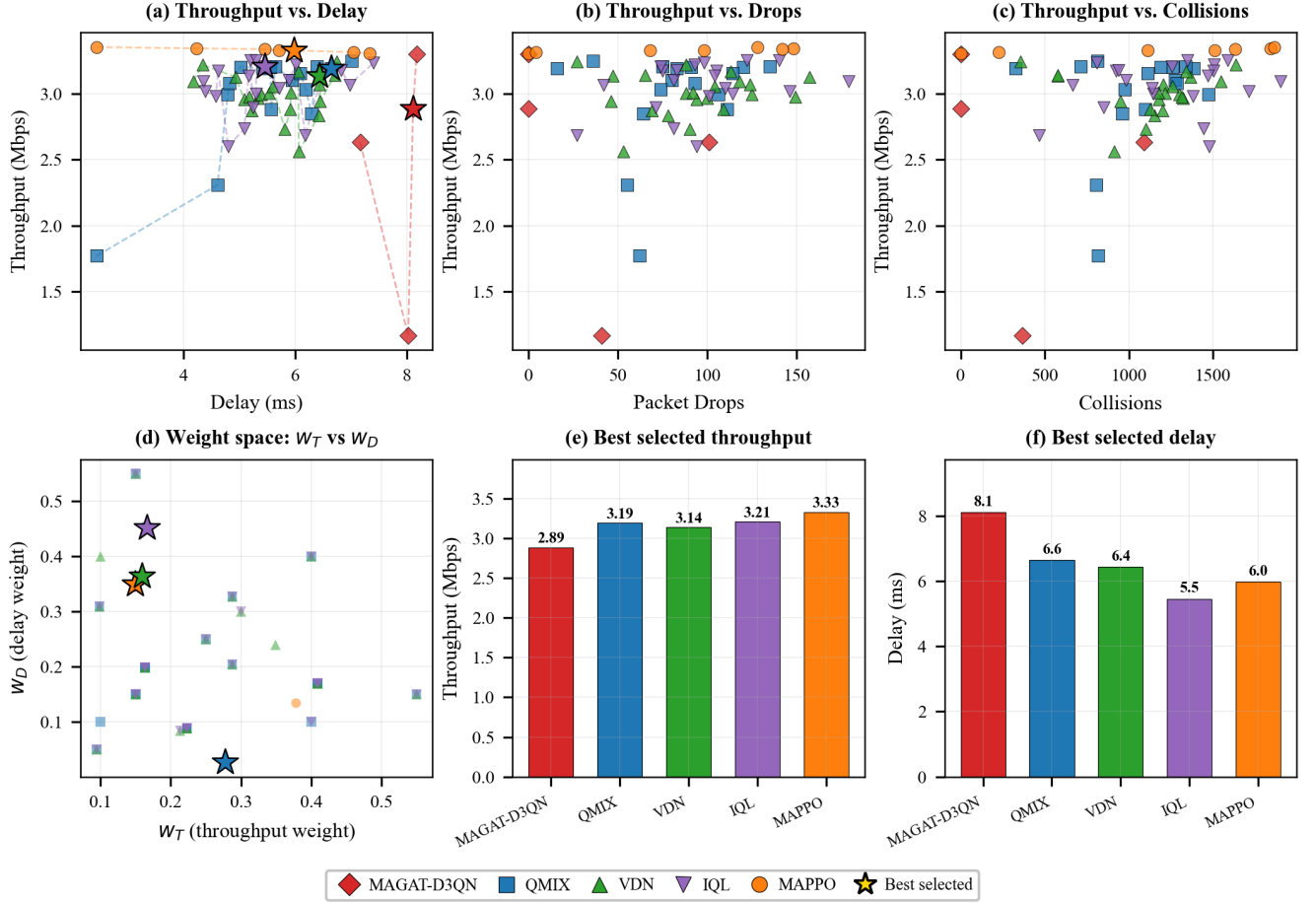


Fig. 5. NSGA-II Pareto tuning analysis over throughput–delay, throughput–drop, and throughput–collision tradeoff planes. The final configuration is selected using the balanced L_2 -to-ideal rule from the nondominated set.

MAGAT-D3QN: $\rho_k(t)$ policy vs. state features (grey = dominant $\rho = 0.8$; coloured = deviation cases)

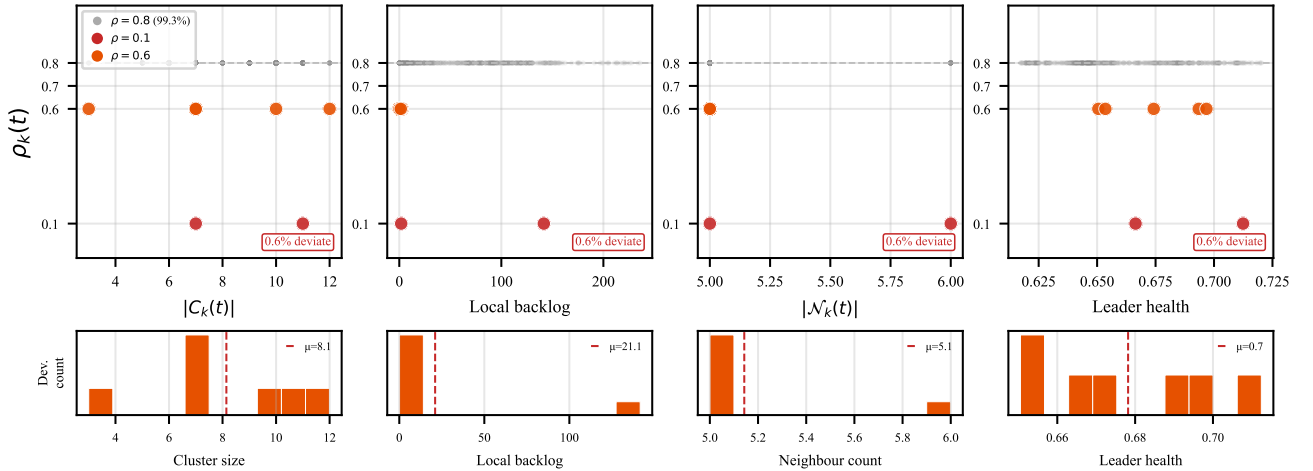


Fig. 6. Learned burst-allocation behavior of MAGAT-D3QN. The selected $\rho_k(t)$ is plotted against cluster size, local backlog, graph degree, and leader-health indicators. The policy concentrates near $\rho_k(t) \approx 0.8$ during stable operation, with adaptive deviations toward lower ρ values under coordination-sensitive conditions.

treats cluster-heads as local decision-making agents over a time-varying inter-cluster interference graph and assigns each cluster-head the burst-level action $a_k(t) = (m_k(t), \rho_k(t))$. This action jointly determines the local access mode and the

split between intra-cluster service and inter-cluster coordination, thereby exposing a service–coordination tradeoff that is not represented by a single global MAC-selection abstraction.

To solve this problem, we developed MAGAT-D3QN, a

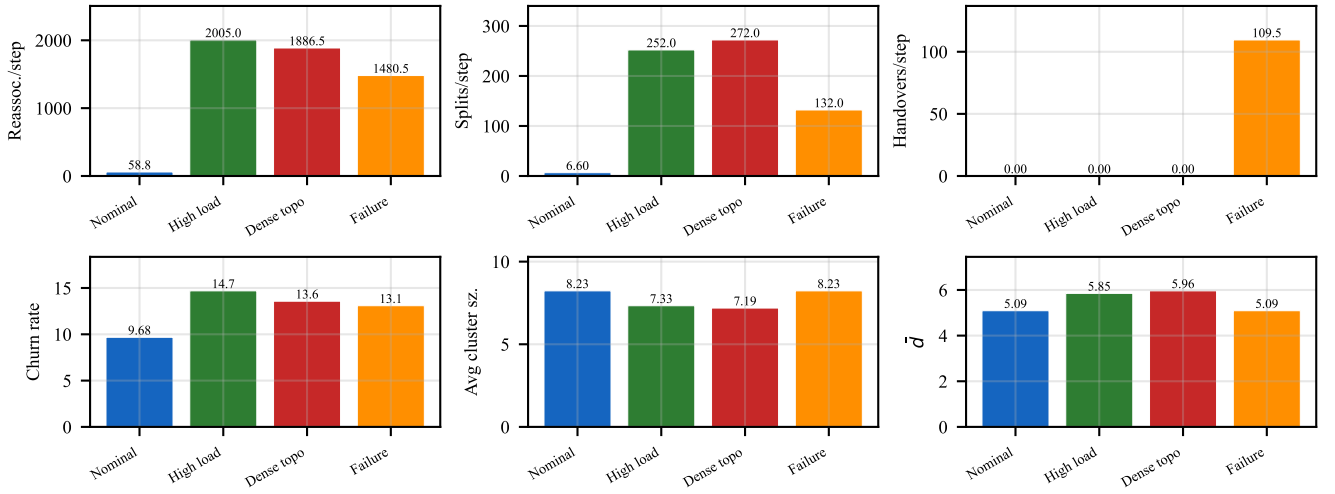


Fig. 7. Cluster-dynamics sanity statistics under nominal load, heavy-tail traffic, compact-dense topology, and failure-window conditions. Reassociation and split counts are episode totals over the 100-step horizon, not per-step rates. Numerical values are reported in Table VI.

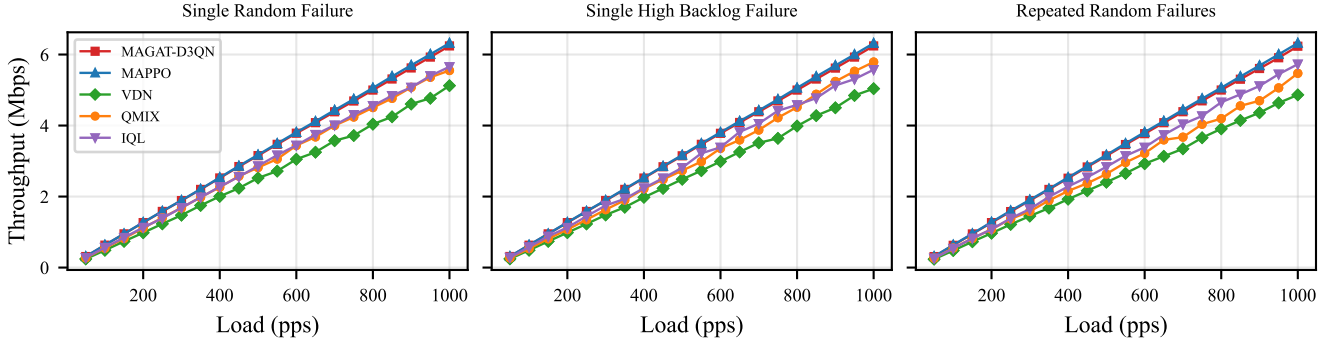


Fig. 8. Leader-failure recovery under single random failure, single high-backlog failure, and repeated random failures. The corresponding throughput and fairness statistics are reported in Table VII.

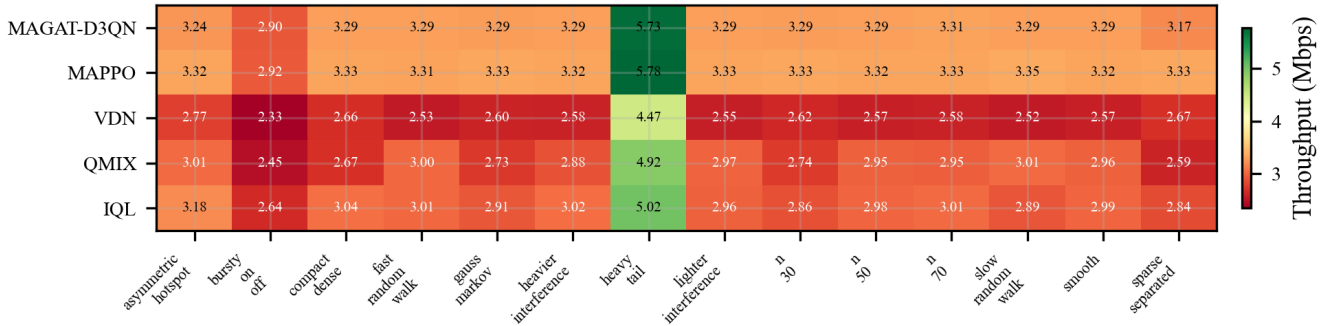


Fig. 9. Generalization under train-test mismatch across node-count, mobility, interference, traffic, and topology regimes. The figure complements the retention statistics in Table ??.

graph-attention recurrent dueling MARL controller for cluster-head-level execution. The evaluation showed that MAGAT-D3QN achieves competitive throughput while eliminating packet drops and collisions at the nominal load. The learned burst-split policy improves throughput over fixed- ρ and heuristic- ρ baselines, and the topology, ablation, failure-recovery, mismatch, and imperfect-information studies indicate that graph representation, recurrent memory, and adaptive burst allocation contribute to coordination stability and fairness-oriented robustness.

The results should be interpreted conservatively. MAGAT-

D3QN does not universally dominate all baselines: MAPPO attains slightly higher raw throughput and IQL attains lower delay in the nominal benchmark. The main value of the proposed framework is instead its reliability-coordination tradeoff and its explicit decentralized formulation for dynamic clustered FANETs. Hardware-oriented profiling further shows that the exported policies can be compiled and executed on representative edge devices, but deployment feasibility remains platform-dependent and requires additional optimization for stricter onboard real-time constraints.

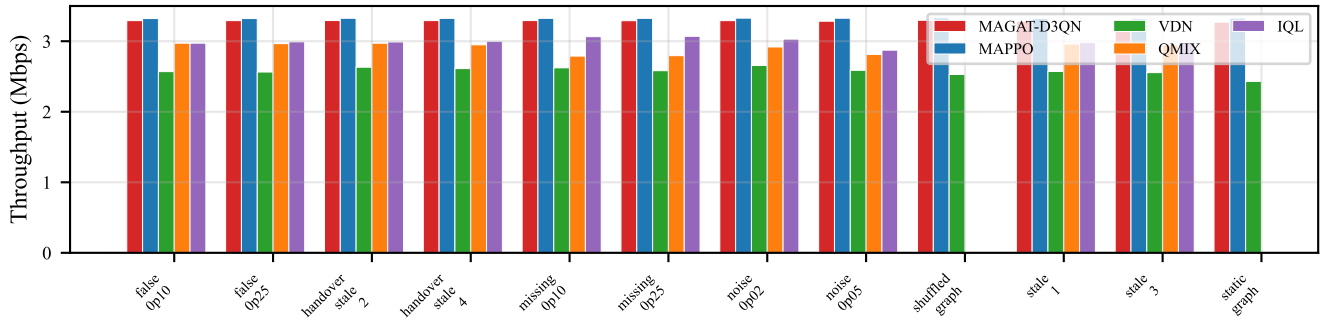


Fig. 10. Robustness to imperfect information, including stale observations, noisy summaries, missing edges, false edges, delayed handover information, and static or shuffled graph variants. Retention statistics are reported in Table VIII.

TABLE III

MAIN OFFERED-LOAD BENCHMARK AT $\lambda = 525$ PPS. VALUES ARE MEAN $\pm$ STD OVER FIVE EVALUATION SEEDS; EXACT ZERO COUNT METRICS ARE REPORTED AS INTEGERS.

Method	$\bar{T}$ Mbps	$\bar{D}$ ms	$\bar{N}_{\text{drop}}$	$\bar{C}$	$\bar{J}$
MAGAT-D3QN	3.31 $\pm$ 0.02	9.29 $\pm$ 0.18	0	0	0.912 $\pm$ 0.008
MAPPO	3.33$\pm$0.01	7.34 $\pm$ 0.21	2 $\pm$ 1	20 $\pm$ 4	0.912 $\pm$ 0.010
VDN	2.59 $\pm$ 0.04	5.62 $\pm$ 0.15	2 $\pm$ 1	23 $\pm$ 5	0.875 $\pm$ 0.012
QMIX	2.97 $\pm$ 0.03	5.57 $\pm$ 0.14	2 $\pm$ 1	29 $\pm$ 6	0.833 $\pm$ 0.015
IQL	2.97 $\pm$ 0.03	5.02$\pm$0.12	0	31 $\pm$ 5	0.822 $\pm$ 0.014

TABLE IV

BURST-SPLIT POLICY COMPARISON AT $\lambda = 525$ PPS.

Policy	$\bar{T}$ Mbps	$\bar{D}$ ms	$\bar{J}$	Coord. ratio	ρ range
Fixed- $\rho = 0.5$	2.07	6.18	0.841	1.00	[0.5, 0.5]
Fixed- $\rho = 0.8$	2.89	8.71	0.873	0.91	[0.8, 0.8]
Heuristic- ρ	2.72	7.51	0.876	1.00	[0.4, 0.7]
Learned- ρ	3.31	9.29	0.920	1.00	[0.10, 0.80]

TABLE V

TOPOLOGY SENSITIVITY AT MATCHED OFFERED LOAD. A: ASYMMETRIC HOTSPOT, B: COMPACT DENSE, C: SPARSE SEPARATED. VALUES ARE MEAN $\pm$ STD OVER FIVE EVALUATION SEEDS.

Top.	Method	$\bar{T}$ (Mbps)	$\bar{D}$ (ms)	$\bar{J}$	Observation
A	MAGAT-D3QN	3.24 $\pm$ 0.02	10.97 $\pm$ 0.22	0.939 $\pm$ 0.009	Graph-aware
A	MAPPO	3.32 $\pm$ 0.01	7.14 $\pm$ 0.19	0.943 $\pm$ 0.007	Consistent
A	VDN	2.77 $\pm$ 0.04	7.91 $\pm$ 0.18	0.917 $\pm$ 0.011	Moderate scaling
A	QMIX	3.01 $\pm$ 0.03	6.91 $\pm$ 0.16	0.879 $\pm$ 0.013	Fairness drop
A	IQL	3.18 $\pm$ 0.03	6.69 $\pm$ 0.15	0.922 $\pm$ 0.010	Local adaptation
B	MAGAT-D3QN	3.29 $\pm$ 0.02	8.07 $\pm$ 0.19	0.887 $\pm$ 0.010	Dense benefit
B	MAPPO	3.33 $\pm$ 0.01	7.08 $\pm$ 0.18	0.886 $\pm$ 0.009	Stable scaling
B	VDN	2.66 $\pm$ 0.04	5.64 $\pm$ 0.14	0.809 $\pm$ 0.015	Interference hit
B	QMIX	2.67 $\pm$ 0.04	5.22 $\pm$ 0.13	0.726 $\pm$ 0.018	Degraded
B	IQL	3.04 $\pm$ 0.03	4.92 $\pm$ 0.12	0.819 $\pm$ 0.014	Fairness issue
C	MAGAT-D3QN	3.17 $\pm$ 0.03	6.65 $\pm$ 0.17	0.848 $\pm$ 0.012	Robust
C	MAPPO	3.33 $\pm$ 0.01	6.56 $\pm$ 0.16	0.859 $\pm$ 0.011	Maintains $\bar{T}$
C	VDN	2.67 $\pm$ 0.04	5.61 $\pm$ 0.14	0.805 $\pm$ 0.016	Range limited
C	QMIX	2.59 $\pm$ 0.04	5.48 $\pm$ 0.13	0.753 $\pm$ 0.019	Weakest
C	IQL	2.83 $\pm$ 0.03	5.98 $\pm$ 0.15	0.827 $\pm$ 0.013	Moderate scaling

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TABLE VI

CLUSTER-DYNAMICS STATISTICS. COUNTS ARE EPISODE TOTALS OVER 100 STEPS.

Scenario	Reassoc.	Splits	Merges	Hand.	Churn	$\bar{d}$
Nominal	58.8	6.6	0.0	0.0	0.012	5.09
Heavy-tail	2005.0	252.0	0.0	0.0	0.401	5.85
Compact-dense	1886.5	272.0	0.0	0.0	0.377	5.96
Failure window	1480.5	132.0	0.0	109.5	0.318	5.09

Note: Average cluster sizes for the four rows are 8.23, 7.33, 7.19, and 8.23, respectively.

TABLE VII

FAILURE-RECOVERY SUMMARY UNDER LEADER FAILURES. THROUGHPUT IS IN MBPS; FAIRNESS IS JAIN'S INDEX.

Method	Rand.		Backlog		Rep. rand.	
	$\bar{T}$	$\bar{J}$	$\bar{T}$	$\bar{J}$	$\bar{T}$	$\bar{J}$
MAGAT-D3QN	3.29 $\pm$ 0.02	0.90	3.29 $\pm$ 0.02	0.90	3.29 $\pm$ 0.03	0.87
MAPPO	3.33 $\pm$ 0.01	0.90	3.32 $\pm$ 0.01	0.89	3.33 $\pm$ 0.01	0.86
VDN	2.64 $\pm$ 0.04	0.88	2.62 $\pm$ 0.05	0.87	2.54 $\pm$ 0.06	0.81
QMIX	2.96 $\pm$ 0.03	0.81	2.95 $\pm$ 0.03	0.80	2.81 $\pm$ 0.05	0.76
IQL	2.99 $\pm$ 0.03	0.82	2.99 $\pm$ 0.03	0.82	2.99 $\pm$ 0.03	0.79

TABLE VIII

ROBUSTNESS TO IMPERFECT GRAPH AND OBSERVATION INFORMATION. ENTRIES REPORT RETENTION IN PERCENT RELATIVE TO THE MATCHED-INFORMATION CASE.

Method	Stale		Noisy		Miss.		False		Delay	
	Th	J	Th	J	Th	J	Th	J	Th	J
MAGAT-D3QN	99	98	99	98	99	98	99	98	99	98
MAPPO	100	98	100	99	100	98	100	98	100	98
VDN	99	97	101	97	101	101	99	97	101	100
QMIX	99	97	96	98	89	95	95	99	100	97
IQL	100	98	99	100	99	105	95	100	101	99

Note: MAGAT denotes MAGAT-D3QN. Th and J denote throughput (Mbps) and Jain-fairness retention (%), respectively. Stale = stale observations, Noisy = noisy summaries, Miss. = missing edges, False = false edges, and Delay = delayed handover information.

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TABLE IX
MAGAT-ISOLATION ABLATION AT $\lambda = 525$ PPS. VALUES ARE
MEAN $\pm$ STD OVER FIVE SEEDS.

Variant	$\bar{T}$	$\bar{D}$	Drops	Colls.	$\bar{J}$
Full MAGAT	3.31 $\pm$ 0.02	9.29 $\pm$ 0.18	0 $\pm$ 0	0 $\pm$ 0	0.912 $\pm$ 0.008
No graph	3.18 $\pm$ 0.04	8.52 $\pm$ 0.24	1 $\pm$ 1	8 $\pm$ 3	0.851 $\pm$ 0.018
No attention	3.24 $\pm$ 0.03	8.89 $\pm$ 0.21	0 $\pm$ 0	3 $\pm$ 2	0.882 $\pm$ 0.013
No GRU	3.20 $\pm$ 0.03	8.74 $\pm$ 0.22	1 $\pm$ 1	5 $\pm$ 3	0.869 $\pm$ 0.015
No burst hist.	3.27 $\pm$ 0.03	9.01 $\pm$ 0.20	0 $\pm$ 0	1 $\pm$ 1	0.895 $\pm$ 0.011
Fixed- ρ	2.89 $\pm$ 0.04	8.71 $\pm$ 0.20	1 $\pm$ 1	4 $\pm$ 2	0.873 $\pm$ 0.014

TABLE X
FADING-MODEL SENSITIVITY: MEAN THROUGHPUT (MBPS) OVER FIVE
SEEDS.

Method	AWGN	Rician	Nakagami	Rayleigh
MAGAT-D3QN	2.96	2.51	2.65	2.45
MAPPO	3.03	2.70	2.54	2.45
VDN	2.30	2.04	2.18	1.88
QMIX	2.82	2.42	2.50	2.24
IQL	2.67	2.50	2.53	2.36
Mean retention	100%	88.3%	90.6%	82.7%

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TABLE XI
COMPLEXITY, SOFTWARE-SIDE INFERENCE, AND HARDWARE-ORIENTED PROFILING SUMMARY. TASK PERFORMANCE IS MEASURED AT THE NOMINAL OFFERED LOAD $\lambda = 525$ PPS.

Model	Task perf.		Footprint				Software latency			Device latency				
	$\bar{T}$ (Mbps)	$\bar{J}$	Params (count)	Param. (M)	Size (MB)	Mem. (MB)	$\bar{\tau}$ (ms)	τ_{95} (ms)	τ_{99} (ms)	CPU(intel xeon) (ms)	i5 (ms)	RPi (ms)	Samsung Glaxy S24 AIHub ms	Dragonwing RB3 Gen 2 AIHub ms
MAGAT-D3QN	3.31	0.912	45,971	0.046	0.184	0.18	2.71	3.15	3.57	2.83	0.17	—	56	876
MAPPO	3.33	0.912	6,930	0.007	0.106	0.06	0.57	0.65	0.99	0.18	0.03	—	50	252
VDN	2.59	0.875	6,930	0.007	0.030	0.03	0.35	0.41	0.59	0.38	0.03	—	52	270
QMIX	2.97	0.833	44,947	0.045	0.179	0.03	0.39	0.57	0.64	4.63	3.06	—	54	1663
IQL	2.97	0.822	2,450	0.002	0.012	0.01	0.39	0.67	0.91	0.21	0.03	—	49	334

Note: $\bar{T}$ and $\bar{J}$ denote nominal throughput and Jain fairness at $\lambda = 525$ pps. Software latency columns report simulator-side mean, 95th-percentile, and 99th-percentile inference latency. CPU denotes local exported-graph CPU inference time. S24 and RB3 denote Qualcomm AI Hub inference latency on Samsung Galaxy S24 Family and Dragonwing RB3 Gen 2 Vision Kit, respectively. Parameter count and exported model size correspond to the profiled inference graphs and may differ from simulator-side training graphs when actor, critic, mixer, or training-only components are included or excluded. Raspberry Pi latency is not yet measured and is reserved for future device-specific profiling.

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